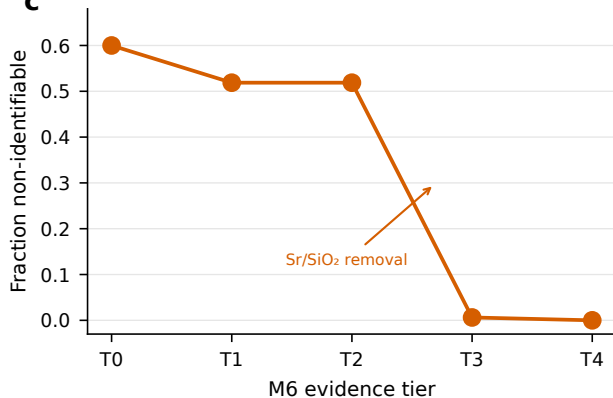


a Observed evidence

✓	Major chemistry
✓	Stable water isotopes
✓	Sr + SiO ₂
✓	Single static-water field
—	Environmental age tracers
—	Screen intervals
—	Repeated heads
—	Independent flow truth

b Defensible field claims

✓	Data readiness / QC
✓	Seasonal chemistry prediction
✓	Reaction equivalence classes
✓	Alternative-edge sensitivity
—	Residence-time validation
—	Exact directed topology
—	Unique field reactions
—	Validated field digital twin

c Evidence-tier limitation**d** Seasonal hold-forward